

Ratish Gupta

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TECHNICAL SKILLS

- **HDL & Verification:** Verilog, SystemVerilog, RTL Design, FSM Design, Datapath & Control Logic, Timing Analysis, Pipelining, cocotb, Verilator
- **Programming:** Python(Numpy, Signal Processing), C/C++, C++17, Java
- **Tools:** Quartus, Modelsim, Questasim, GTKWave, Git, Tcl Scripting (basics), Make
- **Embedded:** ARM Cortex-M, UART, I2C, GPIO, multithreaded systems
- **Debug & Analysis:** Quartus Prime, QuestaSim, ModelSim, Git, Linux, Oscilloscope, Logic Analyzer, LTspice, Pspice, Signal Integrity Analysis

PROJECTS

Parameterized Systolic Matrix Multiplication Unit | *SystemVerilog, cocotb, Verilator, Vivado* Mar 2026 - Apr 2026

- Designed parameterized 8×8 output-stationary INT8/INT32 systolic MAC array in SystemVerilog for transformer Q/K/V/O and FNN matmuls (verified at 4×4 and 8×8 with parameter-only changes).
- Built a cocotb + Verilator verification comparing each PE accumulator with bit-precision against a Numpy reference model with 10,000 random matmuls passed at both array sizes with zero RTL changes.
- Synthesized for Xilinx Artix-7 (XC7A100T) with 64 DSP48E1 slices and closed timing at 100MHz with 3.76ns positive slack (FMax = 160MHz, peak 12.8 GOPS (0.0128 TOPS) at 100 MHz for an 8×8 kernel excluding memory bandwidth)

Real-Time Software-Defined Radio System | *C++, Python* Feb 2026 - Mar 2026

- Implemented a real-time multithreaded FM SDR receiver in C++17 on Raspberry Pi with mono/stereo audio and RDS metadata decoding via a 3-thread producer-consumer pipeline
- Designed polyphase rational FIR resampler supporting arbitrary U/D ratios. eliminated intermediate buffer allocation, yielding measurable latency reduction on embedded hardware convolution by 2.7x
- Prototyped full RDS decode chain in Python (BPF, carrier recovery, PLL, IQ demod, RRC, CDR, Manchester, differential, frame sync), then ported to C++
- Diagnosed and resolved block-boundary CDR alignment bug in streaming clock-and-data recovery, confirmed via IQ constellation visualization

RTL Image Decompression Pipeline | *SystemVerilog, Quartus* Oct 2025 - Nov 2025

- Designed and implemented a full FPGA-based image decoder in SystemVerilog on Altera DE1-SoC at 50 MHz, covering chroma upsampling, YCbCr to RGB Color Space Conversion, and 2D inverse DCT
- Implemented pipelined 8-state FSM for upsampling and color-space conversion with 4 hardware-multiplexed multipliers, processing 2 pixels per 8-clock cycle
- Debugged fixed-point rounding errors, 1-bit address overflow, and simulation-hardware mismatches through ModelSim waveform analysis and Quartus timing reports.
- Achieved 55 MHz Fmax with positive timing slack by optimizing critical paths identified in Quartus timing analysis

EXPERIENCE

Bank of Montreal (BMO) May 2025 - Aug 2025 *Software Developer Intern* Toronto, ON

- Developed and validated backend microservices with emphasis on correctness, structured testing, and failure-mode analysis.
- Designed automated test frameworks raising regression coverage to 80%+, implemented edge-case validation across distribution services.
- Performed root-cause debugging across distributed systems, collaborating with cross-functional teams under production deadlines

EDUCATION

McMaster University 2023 - 2027 *Bachelor, Computer Engineering* Hamilton, ON

- **Achievements:** Dean's Honours List (2024), Engineering Int'l Scholar Award, Faculty of Eng Award of Excellence, Dean's Global Distinction
- **Coursework:** Electronic Devices & Circuits (A), Microprocessor Systems (A), Communication Systems (A+), Signals & Systems (A), Data Structures and Algorithms (A), Control Systems

TECHNICAL CONFERENCES

- **IEEE ISSCC 2026 - San Francisco, CA:** Attended AI accelerator and Compute-in-Memory sessions focused on SRAM constraints, KV-cache scaling, and TOPS/W vs TOPS/mm² tradeoffs. Discussed silicon implementation and system-level design considerations with industry engineers from GSME, Qualcomm, IBM, Samsung, Synopsys etc.